

Industrial Self-Collection App

Complete Mobile & Web Platform Specification

Proposal Date: November 06, 2025

Executive Summary

A comprehensive digital solution automating recyclable material collection from industrial partners. The platform includes 6 distinct user modules with 35+ screens.

PROCESS FLOW (5 MAIN STAGES)

Stage	Actor	Key Actions	System Response
1. Request	Vendor	- Creates vehicle request - Material type & quantity - Pickup date/location	- Request ID generated - Alerts sent to Transporter & POC
2. Approval	Transporter & POC	- Reviews request - Assigns vehicle & driver - Approves/rejects	- Confirmation sent - GPS tracking activated
3. Collection	Vendor Transporter	- Vehicle arrives - Material loaded - Weighing (before & after)	- Gate entry logged - Weight data captured
4. Transport	Transporter	- GPS tracked to BSP - Gate entry at plant - Material unloaded	- Live tracking - Entry/exit logged
5. Closure	Vendor BSP POC	- Final weighing - Invoice created - Payment processed	- Net weight calculated - Receipt generated

USER ROLES, RESPONSIBILITIES & ALL SCREENS

1. VENDOR / INDUSTRIAL PARTNER (10 Screens)

Responsibilities	Access Rights	All Screens
- Create collection requests - Track vehicle status - Acknowledge loading - Manage invoices - View history	- Access own requests only - View assigned vehicle GPS - Create/edit invoices - Upload documents - Receive notifications	1. Login/Registration 2. Vendor Dashboard (summary) 3. Create Vehicle Request 4. My Requests (list view) 5. Request Details & Tracking 6. Loading Acknowledgment 7. Invoice Management 8. Create/Edit Invoice 9. Vendor Profile Settings 10. Notifications Center

2. TRANSPORTER (9 Screens)

Responsibilities	Access Rights	All Screens
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• Accept/reject requests • Assign vehicles & drivers • Fill gate entry forms • Manage fleet • GPS tracking	• View all incoming requests • Assign company resources • Update vehicle status • Access GPS dashboard • Manage drivers	1. Transporter Login 2. Transporter Dashboard 3. Request Management (list) 4. Request Accept/Reject/Reschedule 5. Gate Entry Form 6. Vehicle Management 7. Add/Edit Vehicle 8. Driver Management 9. GPS Tracking Dashboard
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3. BSP POC / OPERATIONS MANAGER (6 Screens)

Responsibilities	Access Rights	All Screens
• Approve/reject requests • Monitor all operations • Generate reports • Day-to-day planning • Performance tracking	• Approve all requests • View all collections • Access full analytics • Reschedule pickups • Export reports	1. POC Login 2. POC Dashboard (metrics) 3. Request Approval Screen 4. Monitoring Dashboard (live) 5. Day-to-Day Planning Calendar 6. Reports & Analytics

4. GATE CLERK (3 Screens)

Responsibilities	Access Rights	All Screens
• Verify vehicle documents • Record gate entries • Link to purchase orders • Track in/out times	• Create gate entries • Verify documents • Assign entry numbers • View active vehicles	1. Gate Entry System 2. Active Vehicles Inside 3. Gate Entry History

5. WEIGHBRIDGE OPERATOR (3 Screens)

Responsibilities	Access Rights	All Screens
• Record vehicle weights • Capture weight slip photos • Calculate net weights • Maintain records	• Enter weight data • Upload images • View weighing history • Generate weight reports	1. Weighing Interface 2. Weighing History 3. Weight Reports

6. SYSTEM ADMINISTRATOR (4 Screens)

Responsibilities	Access Rights	All Screens
• Manage user accounts • Configure system settings • Monitor performance • Generate analytics	• Full system access • User management • System configuration • Audit logs access	1. Admin Dashboard 2. User Management 3. System Configuration 4. Activity Logs & Analytics

TOTAL APPLICATION SCREENS: 35 Screens

Vendor Module: 10 screens • Transporter Module: 9 screens • BSP POC Module: 6 screens
 Gate Clerk Module: 3 screens • Weighbridge Module: 3 screens • Admin Module: 4 screens

Industrial Self-Collection App - Technical & Business Overview

KEY FEATURES & CAPABILITIES

Feature	Description	Business Value
Mobile Apps	Native Android & iOS apps with offline capability	Easy adoption, works without internet
Real-Time GPS	Live vehicle tracking with route history & ETA	Complete transparency, better planning
Automated Workflows	Request → Approval → Collection → Invoice	80% reduction in coordination time
Digital Weighbridge	Electronic weight recording with photo proof	Eliminates disputes, ensures accuracy
Invoice Management	Auto-generated invoices using actual weights	Faster payments, reduced errors
Multi-Channel Alerts	Push, SMS, and email notifications	No missed communications
Document Management	Digital storage of all certificates & receipts	Complete audit trail
Analytics Dashboard	Real-time metrics and performance insights	Data-driven decision making

EXPECTED BENEFITS & ROI

Metric	Current State	With App	Improvement
Coordination Time	2-4 hours	<30 minutes	80% reduction
Vehicle Utilization	60-65%	85-90%	+30% efficiency
Real-Time Visibility	None	100% GPS tracked	Complete transparency
Manual Errors	15-20%	<2%	90% reduction
Documentation	Paper-based	Digital audit trail	100% compliant
Vendor Satisfaction	Moderate	High	Significant improvement

TECHNOLOGY STACK & TIMELINE

Technology Stack	8-Month Implementation Timeline
Mobile: React Native / Flutter (iOS & Android) Backend: Node.js / Python with RESTful APIs Database: PostgreSQL (17 interconnected tables) GPS: Real-time tracking with route optimization Cloud: AWS / Google Cloud with auto-scaling Security: SSL encryption, JWT authentication	Month 1-2: Database setup, APIs & authentication Month 3-4: Mobile apps, request workflows Month 5-6: GPS tracking, weighbridge, invoicing Month 7: Testing & pilot with 2-3 vendors Month 8+: Full rollout, training & support

DATABASE ARCHITECTURE

17 Interconnected Database Tables Organized Into 7 Categories

User Management: users, vendors, transporters • Fleet Management: vehicles, vehicle_types, drivers
Request Processing: vehicle_requests, request_assignments • Materials: material_types
Tracking & Operations: gps_tracking, gate_entries, weighbridge_records • Financial: invoices, loading_acknowledgments
System Support: notifications, documents, activity_logs

NEXT STEPS & ACTION ITEMS

- 1. Review and approve this detailed specification
- 2. Finalize UI/UX preferences and branding requirements
- 3. Conduct stakeholder workshop for screen mockups
- 4. Select 2-3 vendors for pilot testing
- 5. Begin Phase 1 development (Month 1-2: Database & APIs)

This proposal provides complete specifications including 5-stage process flow, 6 user roles with detailed responsibilities and access rights, and all 35 application screens. For additional technical documentation including ERD diagrams and detailed database schema, supplementary documents are available.